



EEE3094S 2024: Exam

- The exam is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write “see attached pages” by the question) and make sure they are marked with your peoplesoft ID.
- There are five questions totalling 103 marks, but the maximum mark is 100.
- You have 15 minutes reading time and 3 hours to complete the exam.
- Information sheets for all questions requiring them are included at the end of the paper. You are advised to remove these for your convenience. If you wish to include calculations or annotations on these sheets in your answer, please reattach them to the paper if possible, or ensure that they are clearly marked with your peoplesoft number.
- The last page is a formula sheet.

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Question 1 [9 marks]

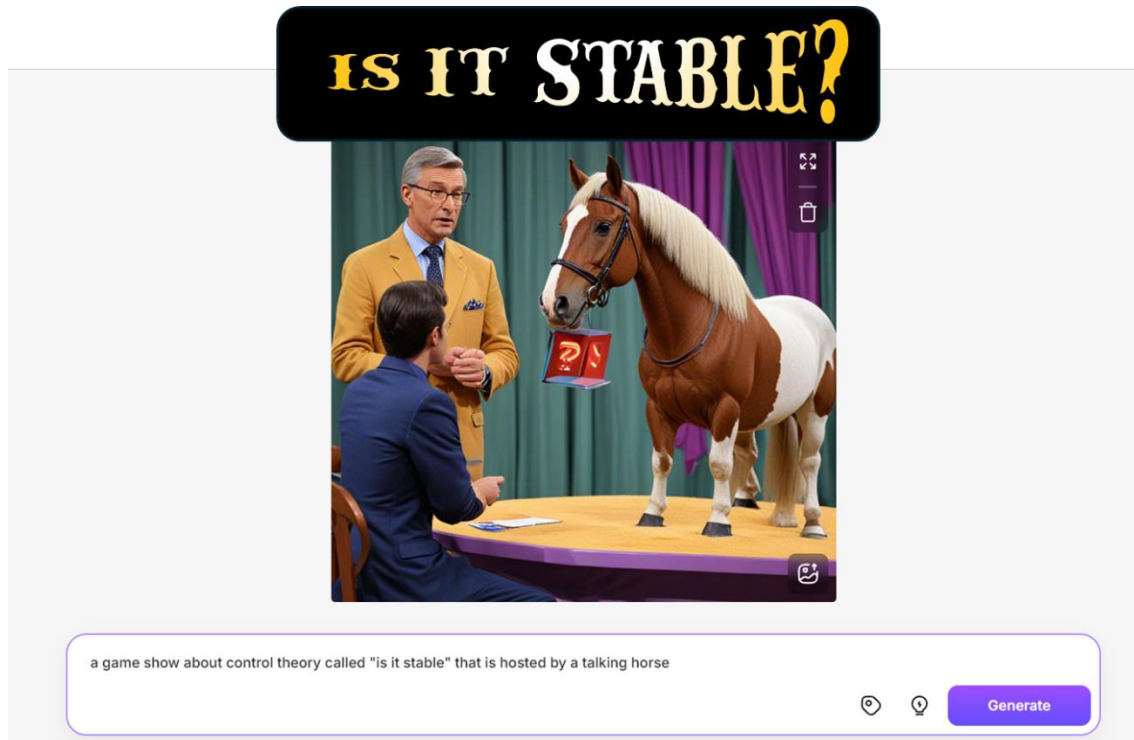
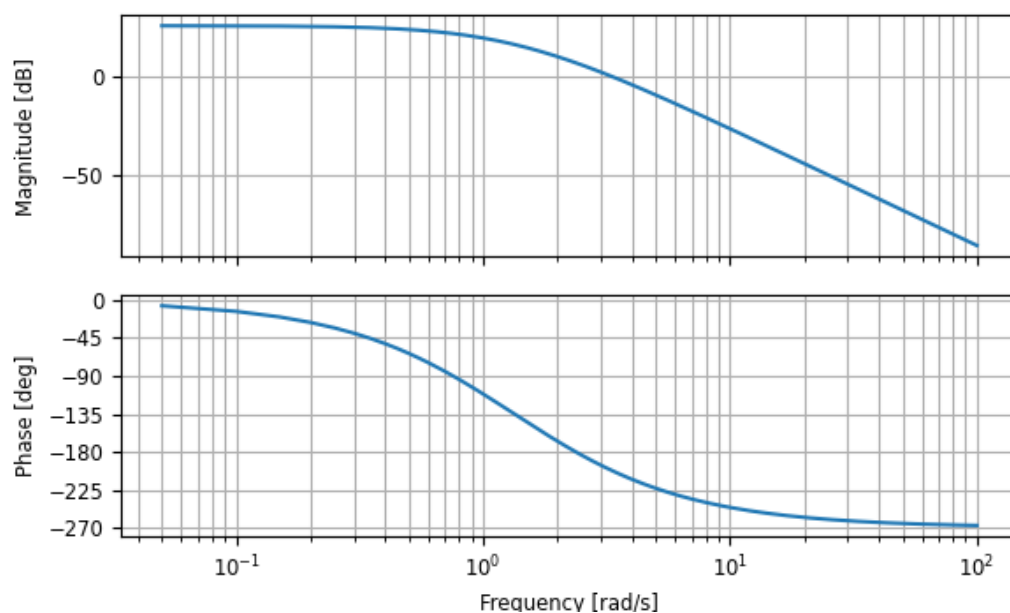


Figure 1: You know what, Midjourney AI? I'll take it.

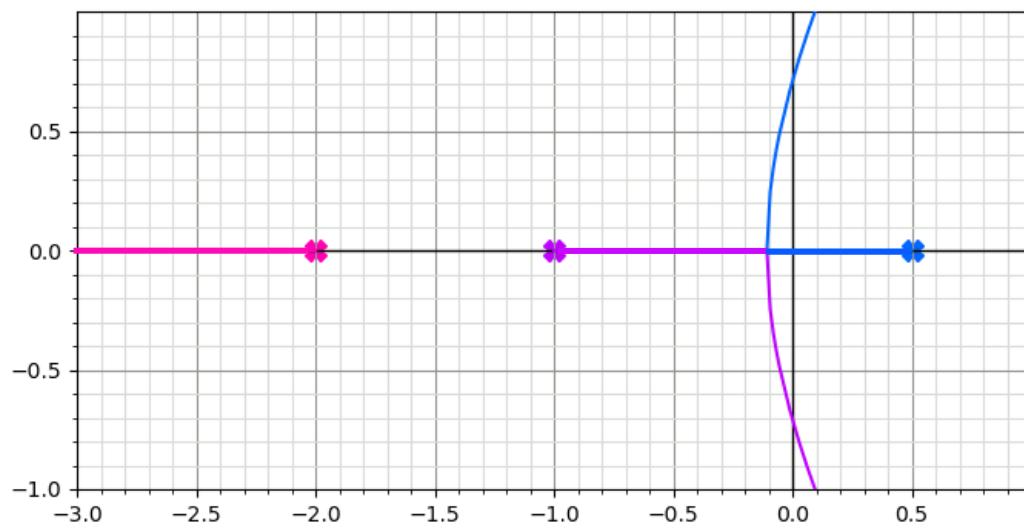
You've heard of Netflix's *Is it Cake?* But what about *Is It Stable?* The game show with way more feedback control theory, and way fewer tasty treats. Here's how it works: you, the contestant must take a look at each of our three eligible systems, and tell us "is it stable?"

Possible answers might be: Yeehaw! (yes) Neigh! (no) Or, it depends! But you need to say what it depends on. For every system you correctly call (provided you give a sensible explanation), you will win 3 marks!

- The closed-loop system, if the open-loop frequency response encircles -1 on the Nyquist plot twice.
- The system with this open-loop frequency response, when put in closed loop:



c) The system with this root locus, when put in closed loop with a proportional controller:

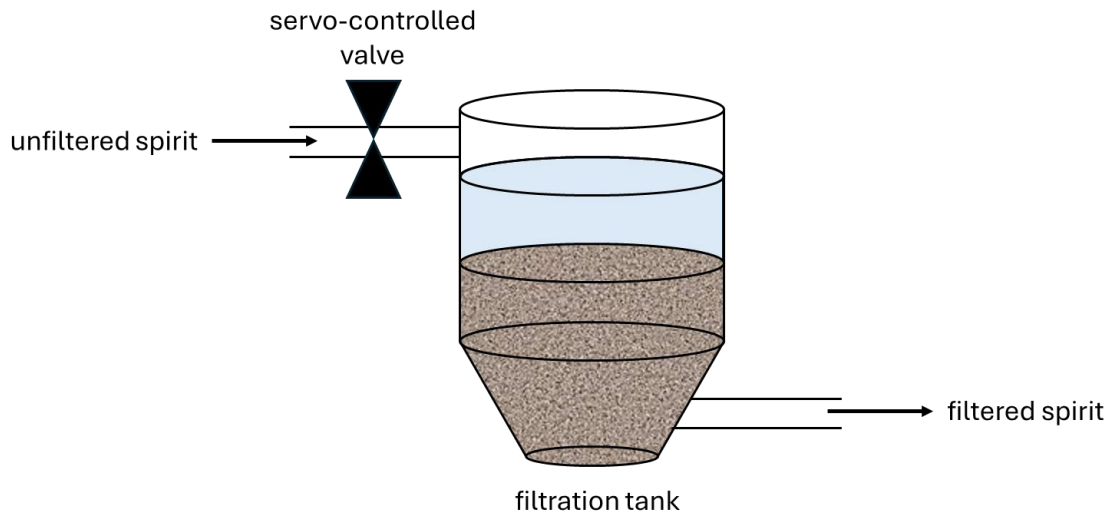


Question 1

Question 1

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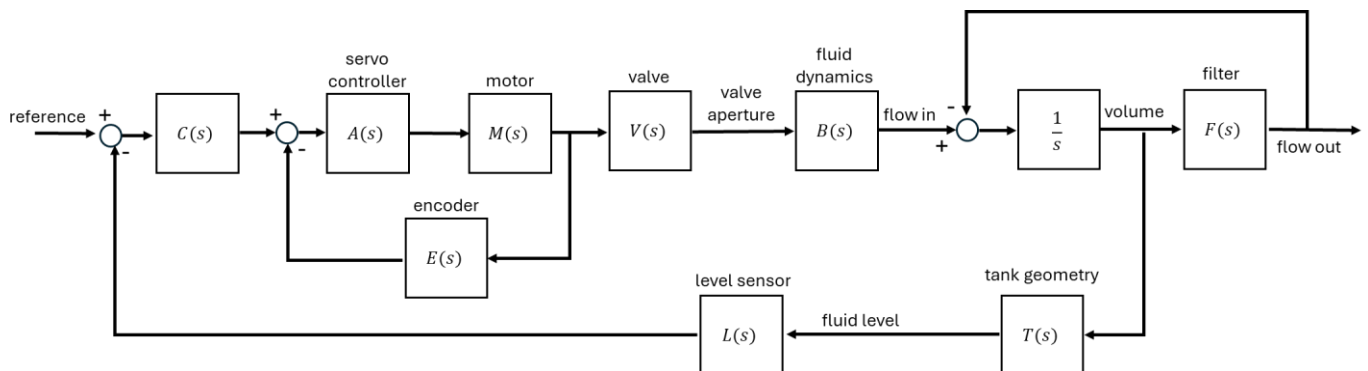
Question 2 [12 marks]



Your friend got into distilling during the quarantine, and this has now turned into a thriving craft spirits business. Believing that you know something about control engineering, they have asked for your help automating their process.

Specifically, they want you to help control the fluid level in their vertical filter. This is a tank containing a carbon filter that the unfiltered spirit is fed into via a variable valve controlled by a servo. Liquid then flows out of the tank at a rate determined by the level in the tank and the properties of the filter. A level sensor monitors the height of the liquid in the tank and feeds it back to the microprocessor controlling the input servo.

A block diagram of this system is shown below:



Simplify the block diagram into a single transfer function. [12 marks]

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Question 2

Question 3 [20 marks]

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The African Robotics Unit wants to create a mobile support frame to help its legged robots stay upright during walking and running tests. Think: the walking rings babies use, but for a robot.

Unlike a walking ring, which the baby must push along, the frame will drive itself on wheels and track the position of the robot as it moves, so it doesn't interfere with the robot's motion.

- a) Using the information given in section A of the information sheet at the end of the paper, create a block diagram model of the mobile support frame. Indicate the signals $x(t)$, $u(t)$, $\tau(t)$, $f(t)$, $d_r(t)$ and $d_m(t)$ in your diagram. [10 marks]

- b) Use your block diagram and section B of the information sheet to derive a transfer function for the open-loop system relating the input voltage $u(t)$ to the position $x(t)$ in terms of the given variables. You may assume that the system is initially at rest. [3 marks]
- c) The parameters of the motor and mass of the frame are given in section C of the information sheet, but you do not know the damping factor μ .

You decide to measure the velocity transient response of the frame by removing the motors, pushing it up to some velocity, and then letting it freewheel until it stops moving. The resulting velocity plot is shown in Section D of the information sheet. Use it to determine μ .

If you did not complete the preceding parts, identify a suitable transfer function model for the given response. [4 marks]

- d) Another engineer in the lab thinks that you can simplify your model by representing the motor using only the gain constant k_m . Do you agree? Explain. [3 marks]



Figure 2: baby in a walking ring

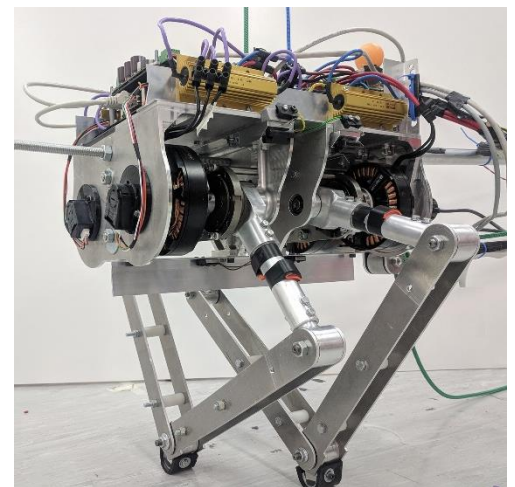


Figure 3: Baleka, the ARU's robot baby (and Gavin's Masters project)

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Question 3

Question 3

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Question 4 [35 marks]

This question follows q3, but can be completed independently.

The support frame is currently being controlled by a proportional controller with a gain of $K = 0.001$. Use the following transfer function for the open-loop support frame (not including the controller):

$$G(s) = \frac{533}{s(16.7s + 1)}$$

The frequency response of the system and controller is shown in various formats on the information sheet at the end of the paper.

- a) Decide whether the following statements about the closed-loop system's tracking ability are TRUE or FALSE. Explain your answers. [4 x 3 marks]
- The frame will be able to match the robot's position perfectly if the robot is standing still.
 - The frame will be able to match the robot's position perfectly if the robot is moving at constant velocity.
 - Some sand gets into the wheel shaft, which increases the damping factor μ . This change can be modelled as a constant offset in the output. It does not affect the frame's ability to track the robot's position.
 - One of the frame's non-driving wheels gets stuck, causing it to slide on the ground without rolling. This can be modelled as a constant disturbing force opposing the frame's motion. It does not affect the frame's ability to track the robot's position.
- b) **Transient Response:** We ideally want the frame to reach the robot's position as quickly as possible, with minimal oscillation (less than 10% overshoot) to prevent unnecessary mechanical wear on the system.
- What is the best achievable settling time (to within 5% of the robot's position) that is possible with proportional control? [3 marks]
 - Roughly sketch the root locus of the system, and indicate the region where the closed-loop poles satisfy the given specifications. [6 marks]
 - Determine the range of gains that satisfy the specifications. Does the given controller gain fall into this range? If not, briefly describe how the system will underperform (i.e. too slow, too much overshoot, both). [5 marks]
- c) **Sensitivity:** We want the system to avoid amplifying disturbances by more than a factor of 2.

We are particularly concerned with periodic disturbances caused by the robot's motion. These will have frequencies matching the robot's gait, which will generally have a frequency of around 6 radians per second. These disturbances must not be amplified by more than 2.5%.

- Indicate these specifications on the appropriate plot following this question. [5 marks]
- Does the given controller gain satisfy these specifications? If not, approximate the largest gain that does, and determine the steady-state error with respect to a ramp input at this value (in metres).
[4 marks]

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Question 4

Question 4

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Question 4

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Question 4 - Plots

Annotate these plots as required for the question. They are also included in the information sheet at the end of the

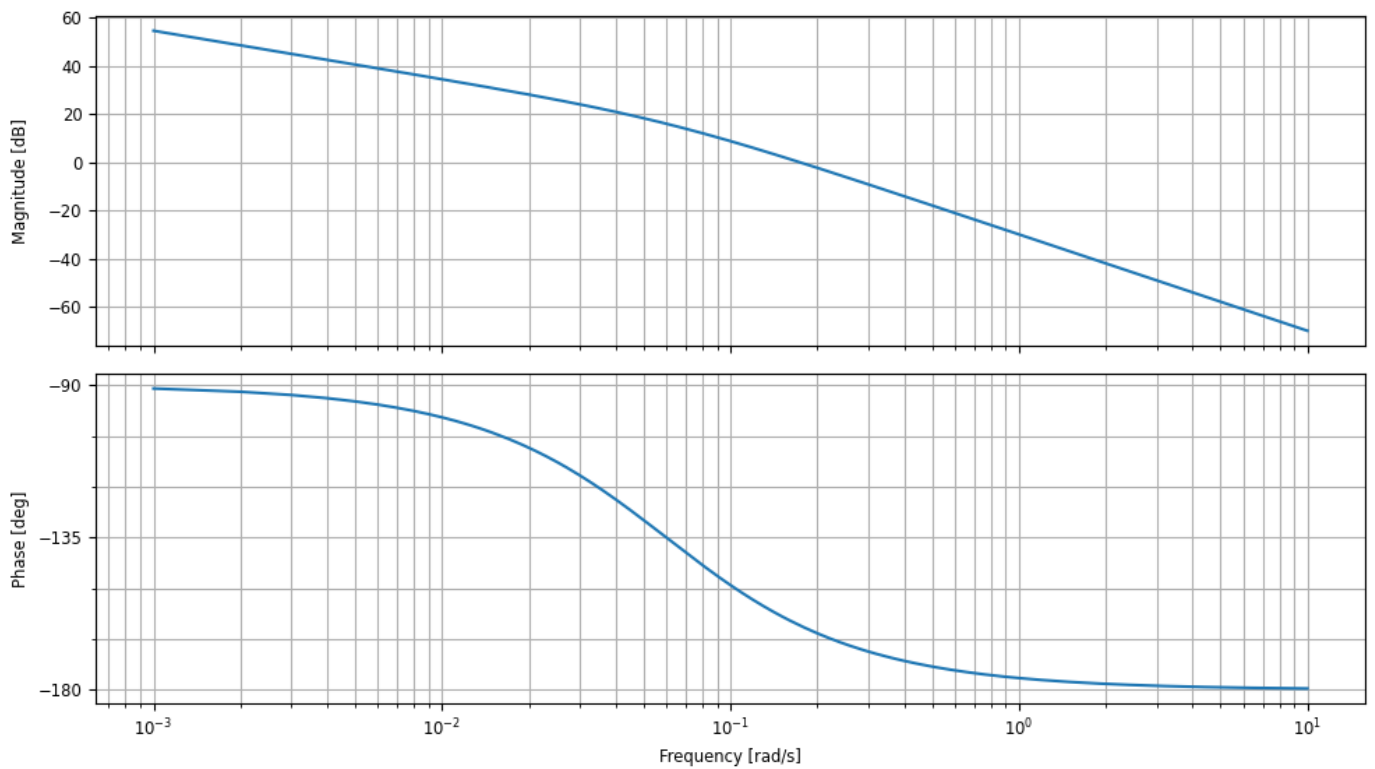


Figure 4: Bode plot of $0.001G(j\omega)$

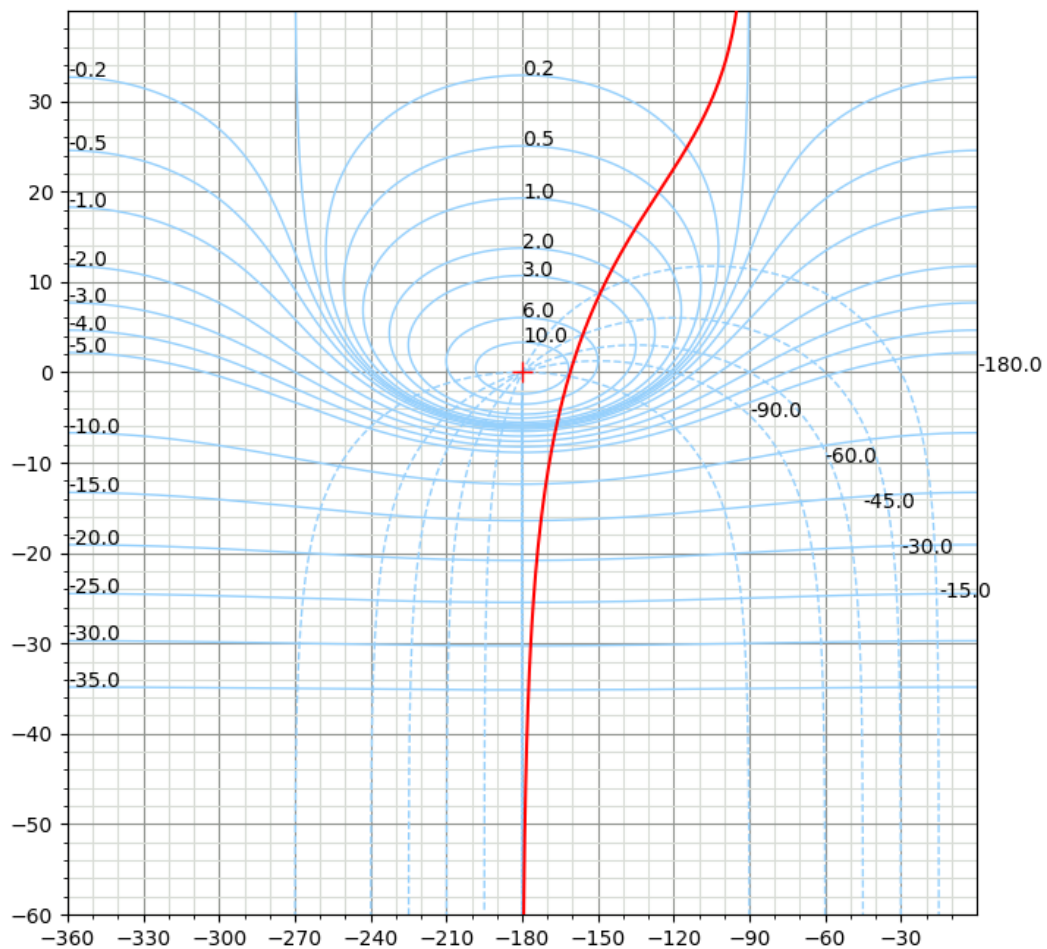


Figure 5: Nichols chart of $0.001G(j\omega)$

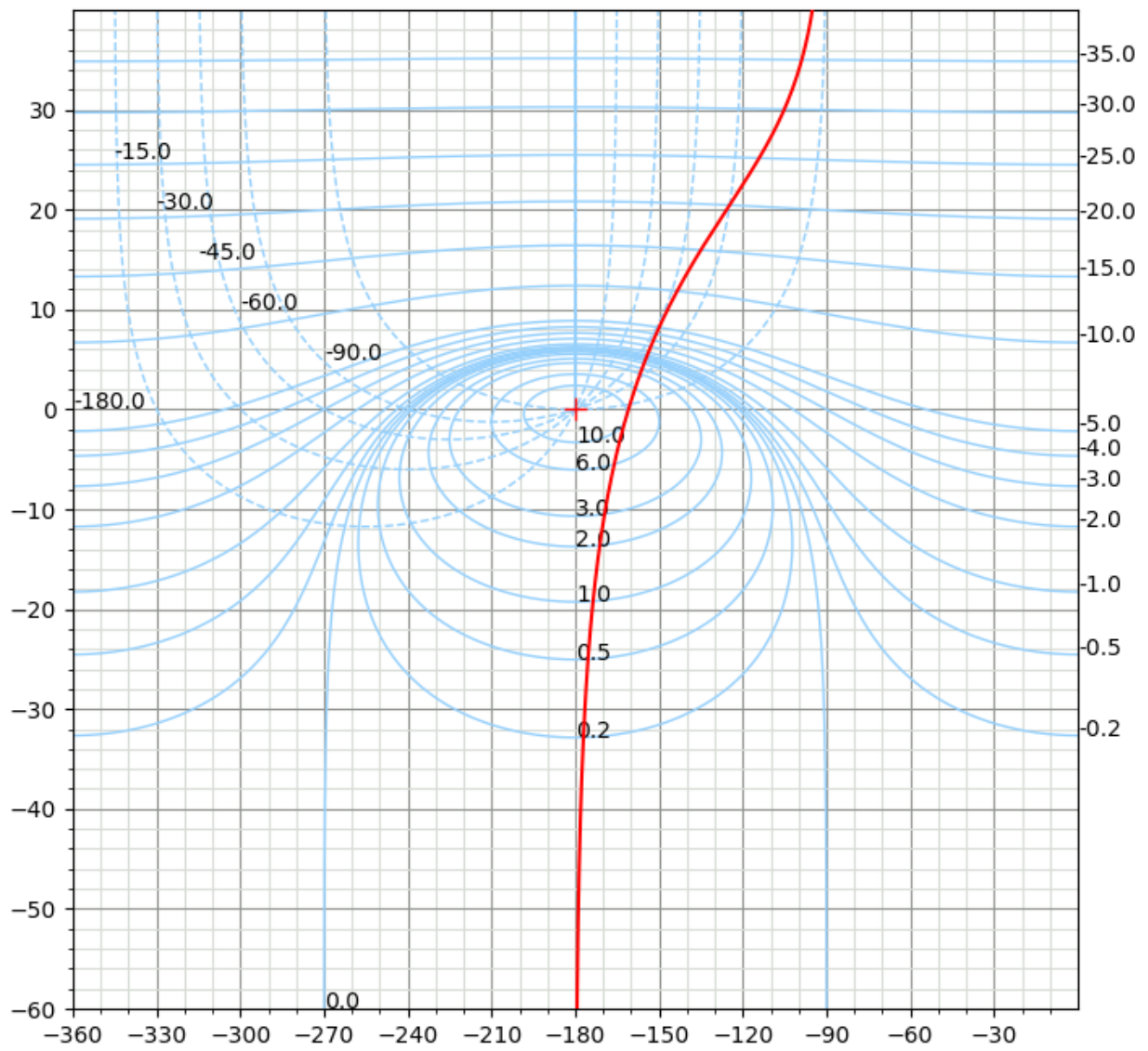


Figure 6: Inverse Nichols chart of $0.001G(j\omega)$

Question 5 [27 marks]

This question follows q3 and q4, but can be completed independently.

The proportional controller's performance is not satisfactory, so you decide to design a more sophisticated controller for the support frame.

- a) The main problem with the proportional controller is that the frame does not track the robot accurately enough. Briefly explain why it is not a good idea to try to fix this problem without changing to a different type of controller. [2 marks]
- b) Another student in the lab suggests that the following controller design would be better:

$$C(s) = \frac{1 \times 10^{-6}}{s}$$

The frequency response plots for the system with this controller are given on an information sheet at the end of the paper.

- i) There is a major problem with this design. What is it? [2 marks]
- ii) You believe that the problem with this controller can be fixed by adding a lead compensator in cascade with it. Design this lead compensator. Your design should be physically realizable and increase the phase margin of the system to at least 30 degrees. It should not affect the low-frequency gain of the system. [8 marks]
- c) Unfortunately, the additional integrator results in a design that is simply too slow, and the frame is left behind by the robot! Design a new lead compensator from scratch for the original system

$$G(s) = \frac{533}{s(16.7s + 1)}$$

You may wish to support your design with a sketch of the root locus.

Aim to place the dominant poles of the system such that the system has a damping ratio of 0.6 and settles to within 5% of its final value in 1 second. (You do not need to account for uncertainty in your design, you can just choose a target pole with these characteristics.)

Clearly describe your design process, and evaluate whether your system is physically realizable, and able to track a ramp input with ≤ 1 cm (0.01 m) of error. You **do not** need to repeat the design if one of these requirements is not met, but you should briefly explain how the design could be changed to do so. [15 marks]

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Question 5

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Question 3 – Information

(For convenience, you may remove this sheet by tearing or cutting along the dotted line.)

Section A – System Overview:

- The frame body P is propelled by two driving wheels, each actuated by an electric motor. The motors are controlled by a microprocessor C , which outputs a voltage $u(t)$ to the motors.
- We will assume that the robot and frame are travelling forward in a straight line, so both motors will receive the same input voltage, and therefore produce the same output torque $\tau(t)$.
- Each motor is connected to the wheel by a gearbox that scales the torque up by a factor α .
- The wheel converts the torque into forward force $f(t)$ acting on the frame. The force produced is equal to the applied torque scaled by r , the radius of the wheel.
- The robot is attached to the frame via bungee cables. You can model the effect of the robot and cables as an unknown disturbing force $d_r(t)$ acting on the frame.
- Friction in the gearboxes and wheel mounts opposes the motion of the frame. This force changes based on factors such as temperature, and how well these components have been lubricated. These fluctuations in the model parameters can be captured as another unknown disturbance $d_m(t)$.
- We want the position of the frame to match the position of the robot. The frame measures its distance from the robot $x(t)$ using a proximity sensor, H . The desired value for $x(t)$ is zero.

Section B – Subsystem models

- The frame is a rigid body with mass m acted upon by two forces: the input force (from the wheels and other external sources) and friction. Friction can be modelled as a simple damping effect that is proportional, but opposite to the velocity of the frame $\dot{x}(t)$, with proportionality constant μ .
- The relationship between the voltage u and output torque τ from the motor is approximated by the transfer function $M(s) = \frac{k_m}{\tau_m s + 1}$, where k_m and τ_m are parameters determined by the motor's electronics and dynamics.
- You can assume that sensor H perfectly measures the output position and transmits it instantaneously to the controller.

Section C – Parameters

Parameter	Value
Frame mass m	5
Damping μ	0.3
Gear ratio α	8
Wheel radius r	0.05
Motor gain k_m	0.5
Motor time constant τ_m	0.0012

Section D - Response Tests

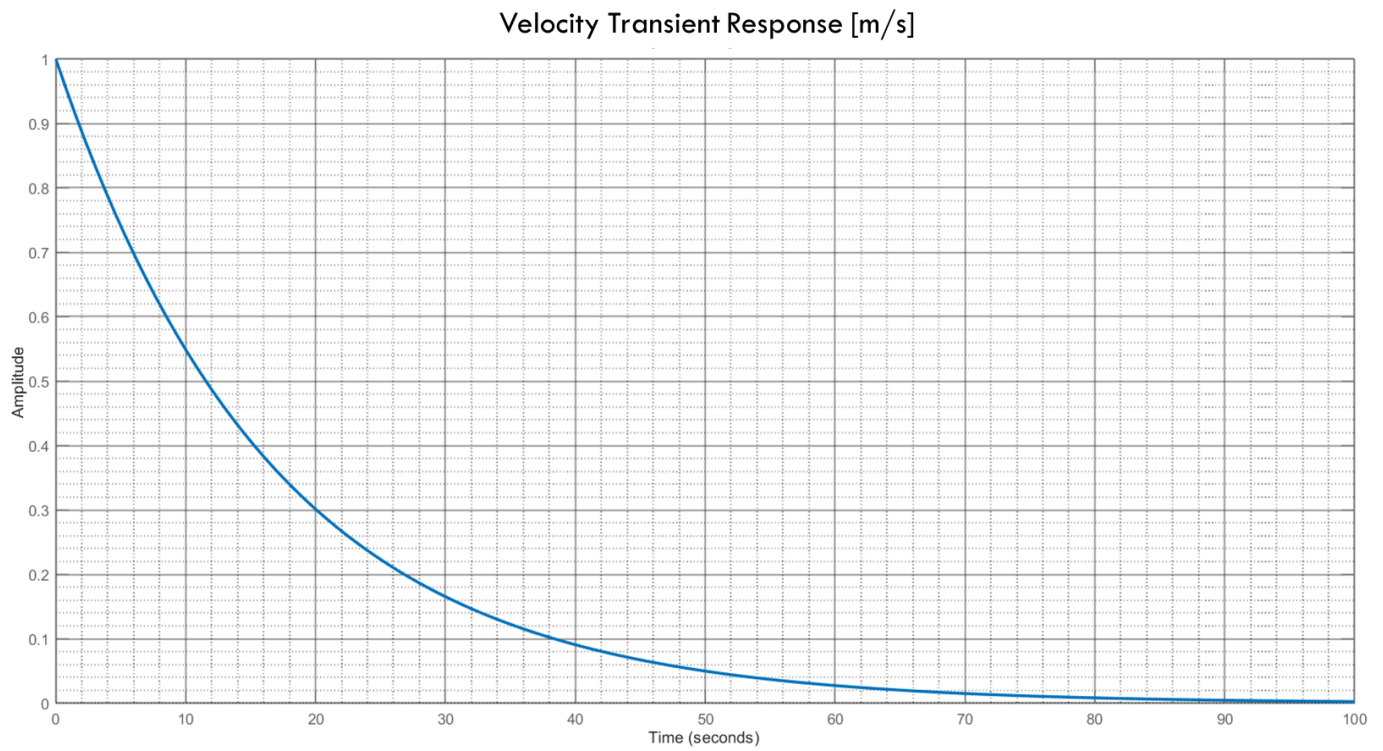


Figure 7: Velocity transient response of support frame with motors removed.

Question 4 – Information

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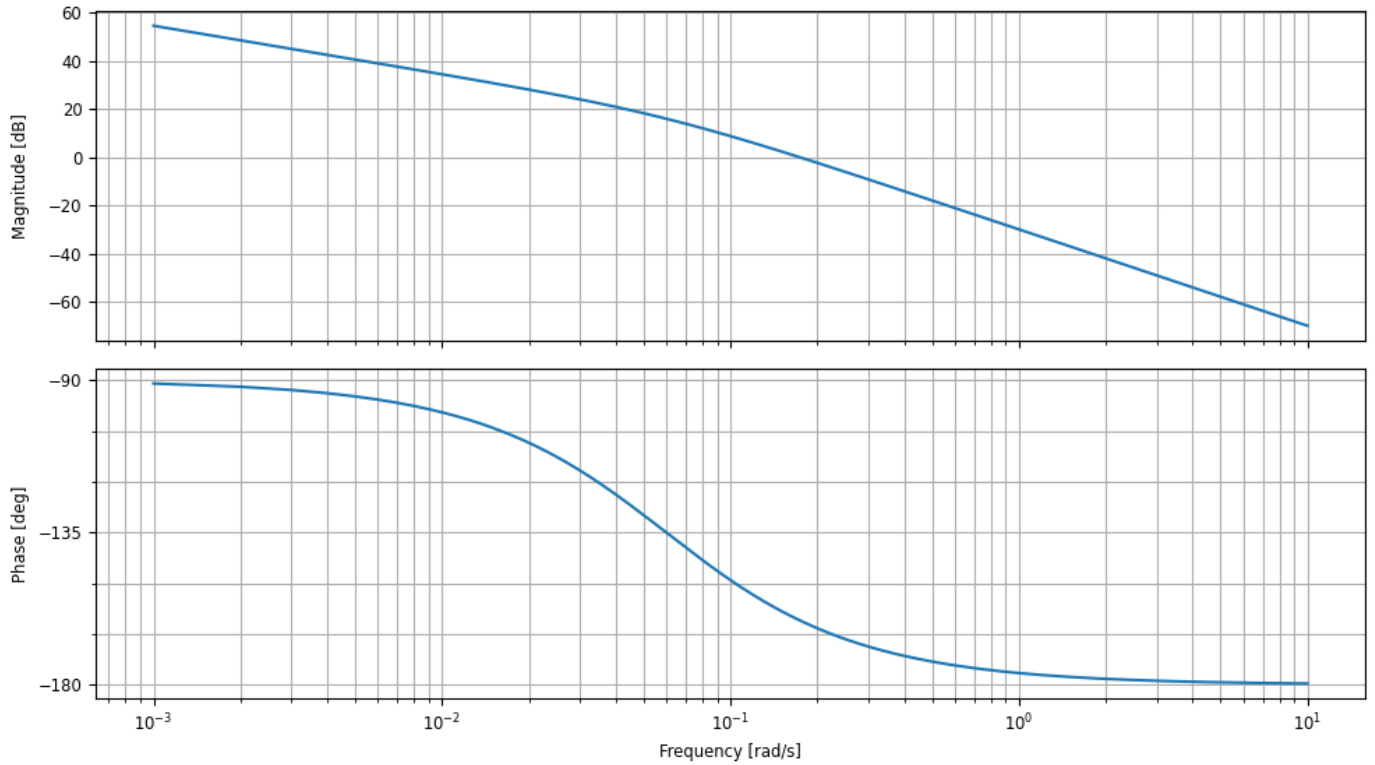


Figure 8: Bode plot of $0.001G(j\omega)$

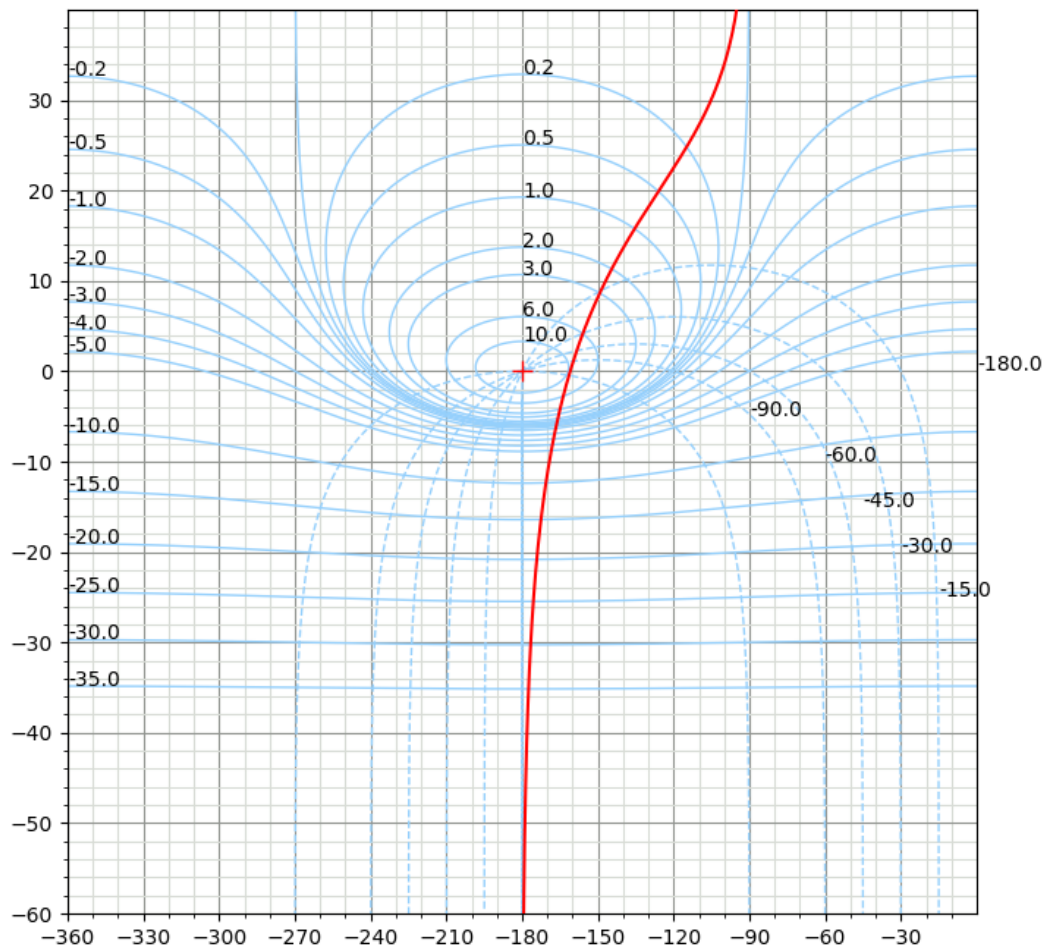


Figure 9: Nichols chart of $0.001G(j\omega)$

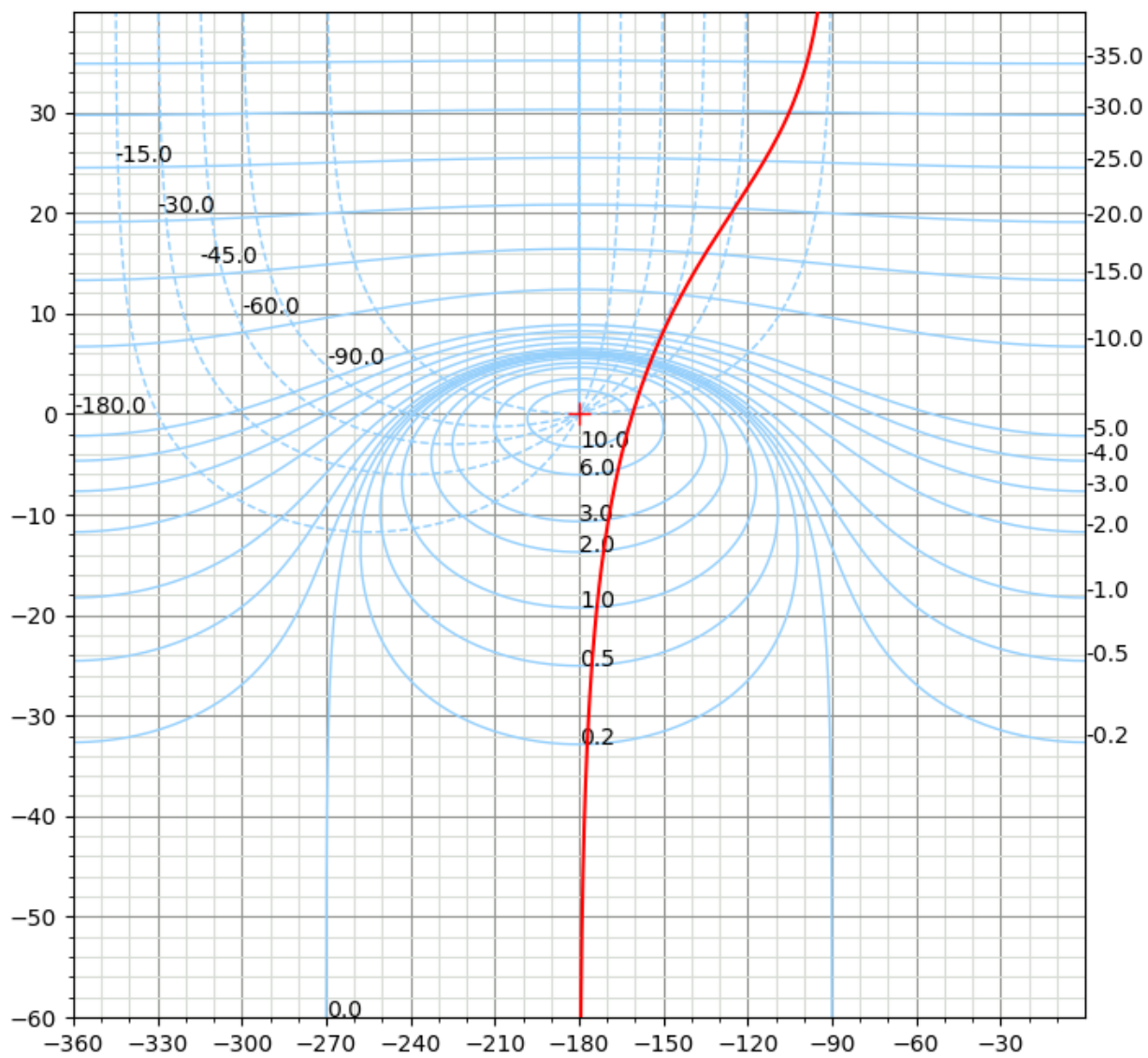


Figure 10: Inverse Nichols chart of $0.001G(j\omega)$

Question 5 – Information

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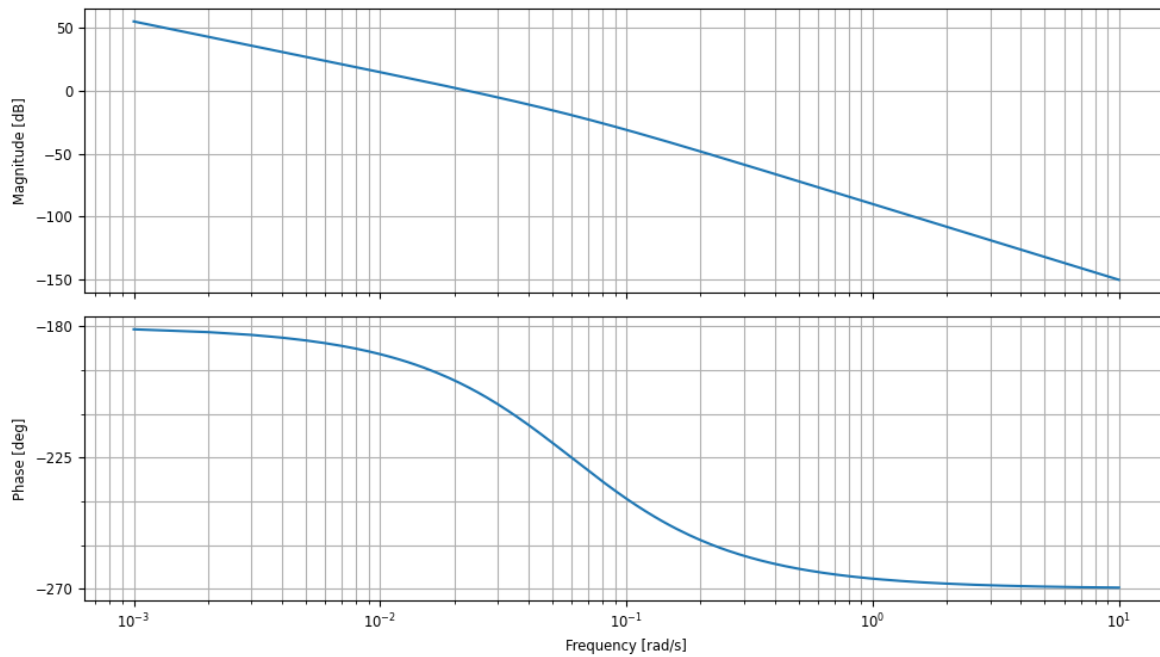


Figure 11: Bode plot of $C(s)G(s)$.

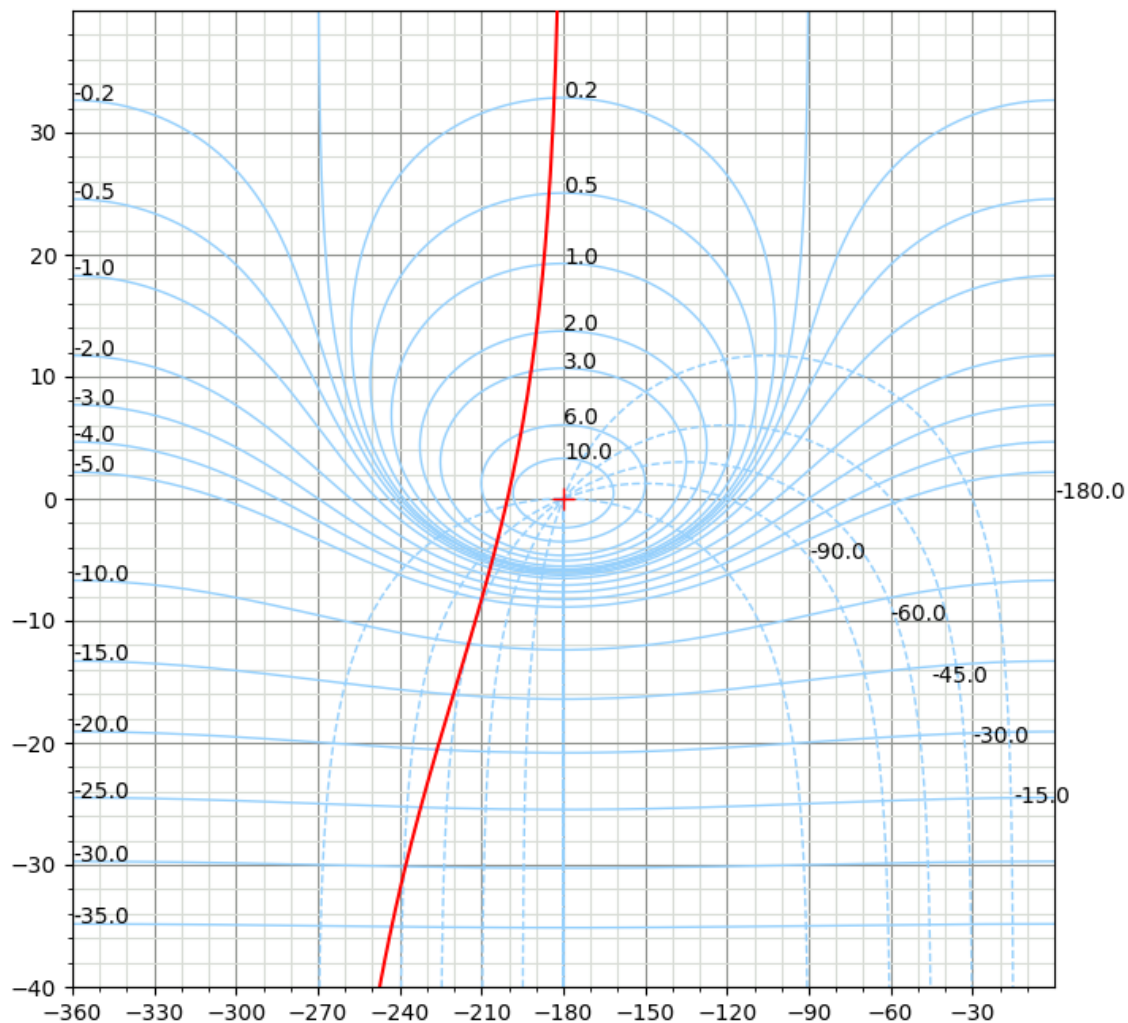


Figure 12: Nichols chart of $C(s)G(s)$

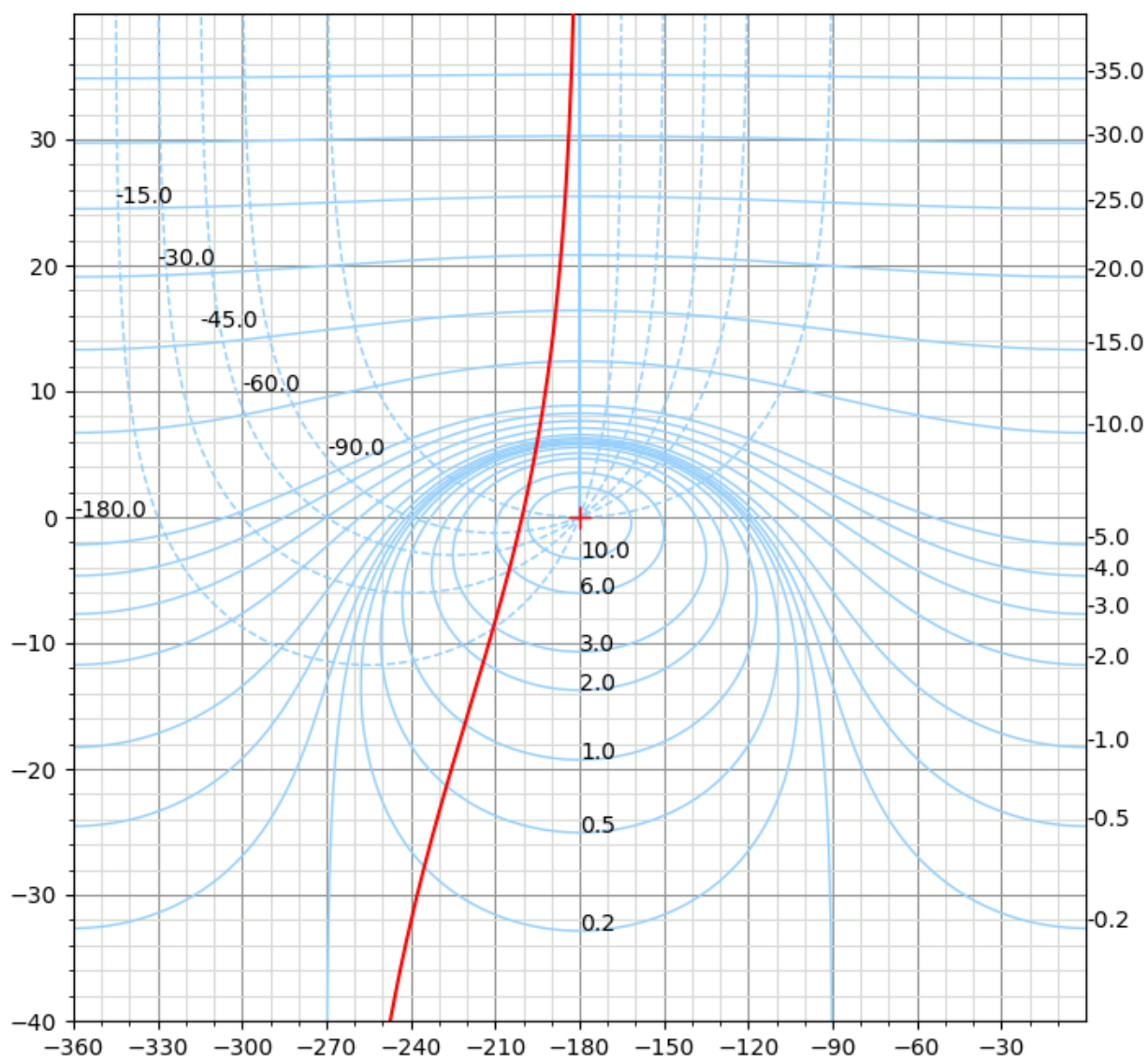


Figure 13: Inverse Nichols chart of $C(s)G(s)$

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\cos \theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A \left(1 + e^{-\zeta\pi / \sqrt{1-\zeta^2}} \right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln \left(\frac{k}{100} \sqrt{1 - \zeta^2} \right)}{-\zeta\omega_n}$$

$$\%OS = 100e^{\left(\frac{-\zeta\pi}{\sqrt{1-\zeta^2}} \right)}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2 + \omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2 + \omega^2}$	<u>cosinusoid</u>
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2 + \omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2 + \omega^2}$	damped <u>cosinusoid</u>